



**UTM**  
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**SECR3553 ARTIFICIAL INTELLIGENCE**

**(KEPINTARAN BUATAN)**

**SECTION 03**

**ASSIGNMENT 1:**

**PARKING DETECTION APPLICATION**

**THEME:**

**SMART CITY**

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## Knowledge Representation

### 1. Detect Occupied Parking Space

IF vehicle\_detected = TRUE THEN space\_status = Occupied

#### Explanation:

When a vehicle is detected in a parking slot through image recognition, the AI system classifies the parking space as occupied. This prevents the system from marking the spot as available to other drivers. The rule ensures data accuracy and helps maintain the system's credibility by providing up-to-date parking information. This KR supports the goal of real-time parking detection and improves the efficiency of managing parking availability in congested areas.

#### Predicate Logic/ First Order Logic:

$$\forall x (\text{ParkingSpace}(x) \wedge \text{VehicleDetected}(x) \rightarrow \text{SpaceStatus}(x, \text{Occupied}))$$

For every parking space, if a vehicle is detected, then the space status is "Occupied."

### 2. Detect Available Parking Space

IF vehicle\_detected = FALSE THEN space\_status = Available

#### Explanation:

If no vehicle is detected in a parking slot, the AI system automatically classifies that slot as "Available." This enables the app to display the space to nearby drivers looking for parking.

#### FOL Representation:

$$\forall x (\text{ParkingSpace}(x) \wedge \neg \text{VehicleDetected}(x) \rightarrow \text{SpaceStatus}(x, \text{Available}))$$

### 3. Provide Nearest Available Parking Spot

IF driver\_request = TRUE THEN display\_nearest(space\_status = Available)

#### Explanation:

When a driver requests to find a parking space, the AI system processes real-time data from the database to identify which spaces are available. It then determines the nearest available spot based on the driver's current location and displays it on the interface. This feature helps drivers save time, reduce frustration, and minimize unnecessary traffic movement within parking areas.

#### FOL Representation:

$$\forall x (\text{DriverRequest}(x) \rightarrow \text{Display}(\text{Nearest}(\text{ParkingSpace}(\text{Available}))))$$

For every driver request, display the nearest available parking space.

#### 4. Update Parking Database

IF space\_status changes THEN update database

##### **Explanation:**

When the status of a parking space changes (for example, from Available to Occupied or vice versa), the system must immediately update the central database to ensure all users see accurate, real-time parking information. This knowledge rule ensures the synchronization between the detection system and the mobile app, maintaining consistency and reliability. It also supports the goal of real-time monitoring and data-driven decision-making for efficient parking management.

##### **FOL Representation:**

$\forall x (\text{ParkingSpace}(x) \wedge \text{StatusChanged}(x) \rightarrow \text{UpdateDatabase}(x))$

For every parking space, if a status change is detected, the system updates the database with the new information.

5.

## **Knowledge Representation Involved to Achieve Goals**

The AI Smart Parking Detection System's Knowledge Representations (KRs) work together to accomplish the project's primary objectives, which include effective parking detection, shorter search times, less traffic and improved user experience. Every KR has a distinct function in facilitating data accuracy, real-time updates and intelligent reasoning.

### **Goal 1: Efficient Parking Detection**

Related KRs:

*KR1: Detect Occupied Parking Space*

*KR2: Detect Available Parking Space*

Explanation:

Using image recognition, these KRs allow the AI system to precisely determine each parking spot's condition. The area is indicated as "Occupied" when a car is spotted and as "Available" otherwise. The system keeps an accurate and current record of parking space availability thanks to its logical structure.

### **Goal 2: Reduced Search Time**

Related KRs:

*KR3: Provide Nearest Available Parking Spot*

Explanation:

This KR enables the system to evaluate every place that is available and suggest the one that is closest to the driver's current location. The AI helps drivers save time and minimizes needless driving around the parking lot by automatically displaying the nearest available space.

### **Goal 3: Real-Time Information and Data Consistency**

Related KRs:

*KR4: Parking Database Update*

Explanation:

This KR makes sure that the database is updated right away whenever a parking space's occupancy status changes. In order to meet the objective of real-time parking detection and enhance overall system reliability, this keeps the detection cameras and the mobile app in sync.

#### **Goal 4: Enhanced User Experience**

Related KRs:

*KR3: Provide Nearest Available Parking Spot*

*KR4: Parking Database Update*

Explanation:

These KRs guarantee that users get precise, up-to-date information and practical directions to the closest parking space. The smooth integration of navigation, data update and detection improves user pleasure and fosters system confidence.

When combined, these Knowledge Representations give the AI system the ability to reason logically, make precise judgments and behave intelligently in order to accomplish its primary goals, which include effective detection, less traffic and a more enjoyable driving experience for all users.